

COURSE TITLE : PRODUCTION/OPERATIONS MANAGEMENT
COURSE CODE : 2254
SEMESTER : 2
COURSE CATEGORY : B
PERIODS/WEEK : 6 (4+2)
PERIODS/SEMESTER : 90
CREDITS : 5

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	INTRODUCTION TO PRODUCTION/OPERATIONS MANAGEMENT -PROCESS PLANNING AND WORK STUDY	25
2	PRODUCTION PLANNING, SCHEDULING,CAPACITY, DECISION	25
3	INVENTORY CONTROL	20
4	PROJECT SCHEDULING AND MANAGEMENT	20

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able :
1	1	To understand the concept of production/operations management
	2	To know operation strategy, plant layout, process planning
2	1	To understand production/planning hierarchy
	2	To now the procedure for production scheduling
3	1	To know different types of inventories and its functions
	2	To know procurement and its principles
4	1	To understand CPM and PERT and its applications
	2	To understand the concept of project management and its characteristics

SPECIFIC OUTCOMES

- 1.1.0 Understand production/operation management
- 1.1.1 Distinguish between production/operation management
- 1.1.2 Explain functions of production/operation management
- 1.1.3 Explain the need for better business strategy
- 1.2.0 Understand operation strategy
- 1.2.1 Define operation strategy
- 1.2.2 Explain elements of operation strategy
- 1.2.3 Explain the need for forecasting in production/operation management
- 1.2.4 Explain the different types of production system
- 1.3.0 Understand plant layout
- 1.3.1 Explain plant layout
- 1.3.2 Explain different types of plant layout
- 1.4.0 Understand process planning
- 1.4.1 Define process planning
- 1.4.2 Explain process charts
- 1.4.3 Different types of process charts

- 1.5.0 Understand work measurement
- 1.5.1 Define work measurement
- 1.5.2 Explain work study
- 1.5.3 Explain method study
- 1.5.4 Explain Therbligs

- 2.1.0 Understand production planning hierarchy
- 2.1.1 Explain sales forecasting
- 2.1.2 Explain aggregate planning
- 2.1.3 Explain aggregate demand
- 2.1.4 Describe aggregate plans for services
- 2.1.5 Explain mathematical models for aggregate planning
- 2.2.0 Understand master production scheduling
- 2.2.1 Explain master production scheduling
- 2.2.2 Explain procedures for developing master production schedules
- 2.2.3 Describe MP's in produce to stock and produce to order for firms
- 2.3.1 Define capacity

- 3.1.0 Explain different types of inventory control system
- 3.1.1 Explain ABC technique of Inventory Control
- 3.1.2 Explain VED analysis
- 3.1.3 Explain SDE analysis
- 3.1.4 Explain HML, FSN analysis

- 3.2.0 Understand procurement
- 3.2.1 Describe procurement
- 3.2.2 Explain procurement procedure
- 3.2.3 Explain centralized and decentralized purchasing
- 3.2.4 Explain various methods of purchasing
- 3.2.5 Explain five 'R' principles of purchasing
- 3.3.0 Understand value analysis
- 3.3.1 Explain value analysis
- 3.3.2 Explain codification
- 3.3.3 Explain simplification

- 4.1.0 Understand Project scheduling with CPM/PERT
- 4.1.1 Explain Objectives of Network analysis
- 4.1.2 Describe concepts in net work
- 4.1.3 Explain Critical path method
- 4.1.4 Explain Programme evaluation and review technique

- 4.2.0 Understand Project management
- 4.2.1 Define Project management concepts
- 4.2.2 Describe characteristics of project
- 4.2.3 Explain Project identification
- 4.2.4 Explain sources of project ideas
- 4.2.5 List Project ideas of existing industries

CONTENT DETAILS

- Module I: Production/operation management – meaning comparison – functions – need – operations strategy – elements – need for forecasting – types of production system – facility location – meaning – plant layout – types of plant layout. Process planning – meaning – procedure – factors affecting – process chart – types – work measurement – meaning – work study – Therbligs.
- Module II: Production planning Hierarchy – sales forecasting – meaning – aggregate planning – aggregate demand – aggregate plans for services – mathematical models – MP's in produce to stock to order —control – control charts — capacity decisions – strategies for capacity planning.
- Module III: Different types of Inventory Control System-ABC – VED – SDE – HML – FSN – procurement – meaning – purchasing methods – five 'R' principles – value analysis – meaning – codification – simplification.
- Module IV: Network analysis-Critical path method-Program evaluation and review technique- Project management - characteristics of project-Project identification-sources of project ideas- Project ideas of existing Industries.

BOOKS RECOMMENDED

1. Norman Gaither, Production/Operations Management, Saunders college, publication
2. Aswathappa , Production/Operations Management, Himalaya publishing

BOOKS FOR FURTHER REFERENCES

1. Levin & Others, Production/operations Management, McGraw Hill
2. Martand Telsang, Industrial Engineering & Production Management, S.Chand
3. Chidambara Natha, Engineering Management, PHI